KINGSCLIFF HIGH SCHOOL

ACHIEVEMENT THROUGH ENDEAVOUR

TERM 4

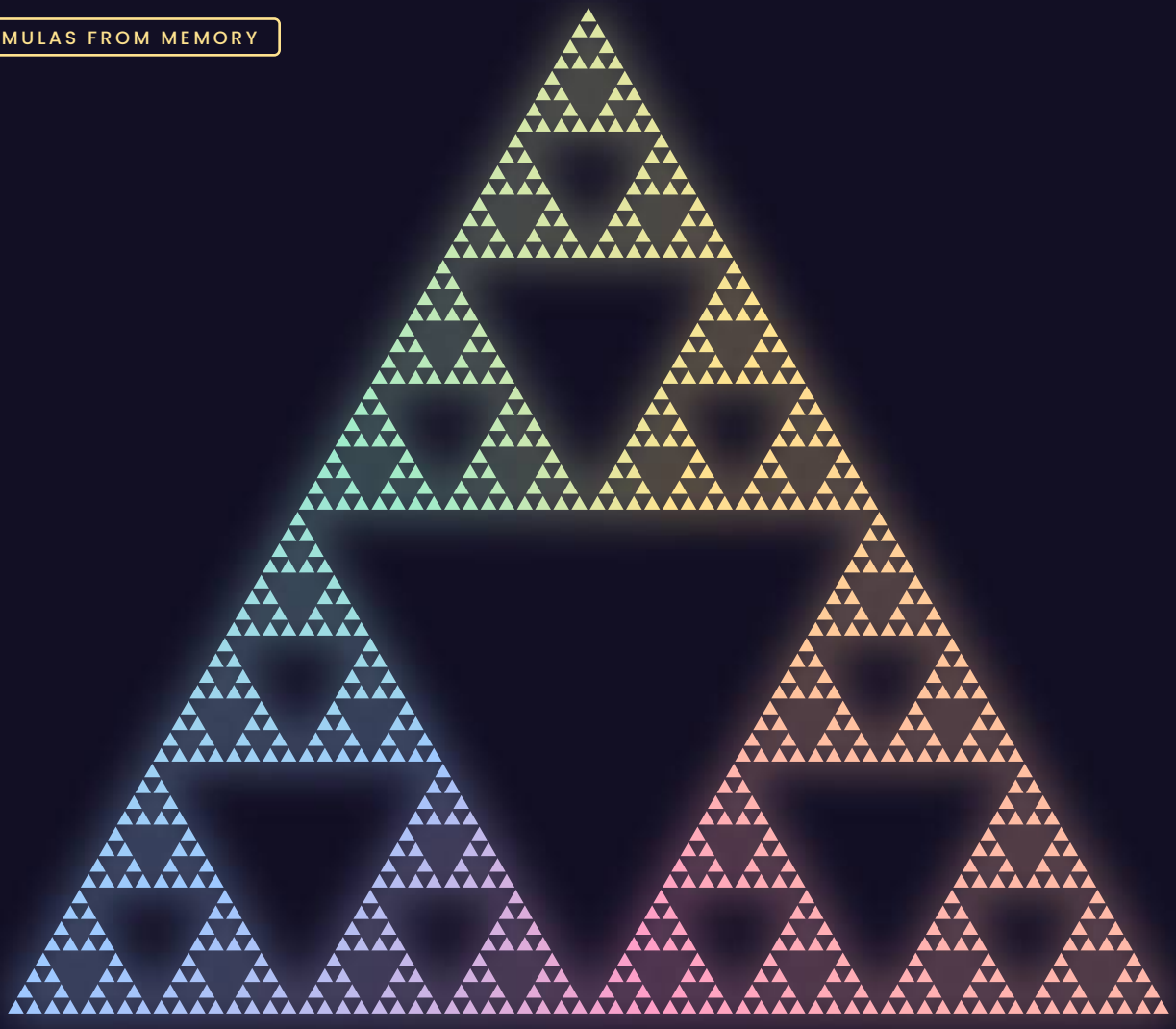
FIVE QUESTIONS. EVERY LESSON.

The Fractal Five

YEAR 10

Weeks 1–10 · 30 sessions · 150 questions

FORMULAS FROM MEMORY



NAME _____

CLASS _____

SIERPIŃSKI TRIANGLE
three copies of itself, each half the size

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

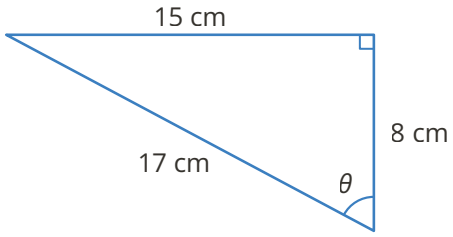
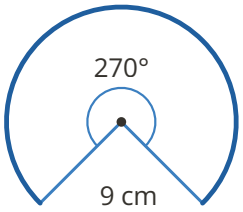
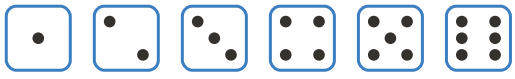
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 1 · Session 1

TERM 4 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Write $\sin\theta$ as a fraction.  	FORMULA SUBSTITUTE ANSWER
2 Find the arc length, correct to 1 decimal place.  	FORMULA SUBSTITUTE ANSWER
3 Mia earns \$30 an hour and works 15 hours. How much does Mia earn?	FORMULA SUBSTITUTE ANSWER
4 Find v when $u = 5$, $a = 3$ and $t = 4$. $v = u + at$	FORMULA SUBSTITUTE ANSWER
5 A fair die is rolled. Find the probability of rolling an even number. 	FORMULA SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

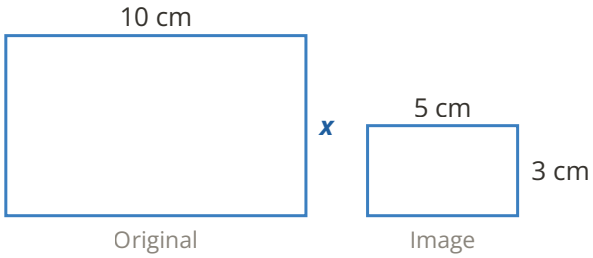
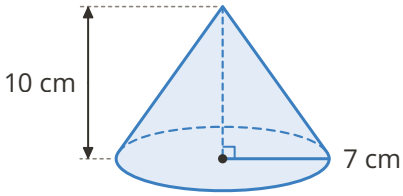
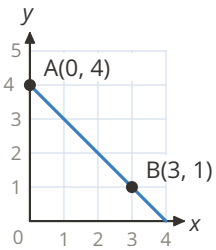
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 1 · Session 2

TERM 4 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 These rectangles are similar. Find x.  	FORMULA SUBSTITUTE ANSWER
2 Find the volume of this cone, correct to 1 decimal place.  	FORMULA $V = \frac{1}{3}\pi r^2 h$ COPY SUBSTITUTE ANSWER
3 Find the percentage profit. Cost price \$30 Selling price \$45	FORMULA SUBSTITUTE ANSWER
4 Write in scientific notation. 820	FORMULA SUBSTITUTE ANSWER
5 Find the gradient of line AB.  	FORMULA SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

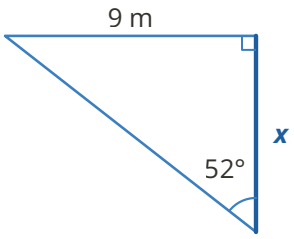
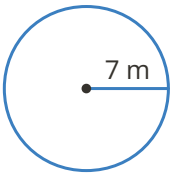
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 1 · Session 3

TERM 4 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find x, correct to 1 decimal place.  	FORMULA SUBSTITUTE ANSWER
2 Find the area of this circle, correct to 1 decimal place.  	FORMULA SUBSTITUTE ANSWER
3 Ella earns a salary of \$68 500 a year. Find Ella's weekly pay, to the nearest cent.	FORMULA SUBSTITUTE ANSWER
4 Evaluate. 5×10^{-2}	FORMULA SUBSTITUTE ANSWER
5 Find the mean of these scores, correct to 1 decimal place. Score: 10, 11, 12, 13, 14 Frequency: 2, 6, 4, 5, 1	FORMULA SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

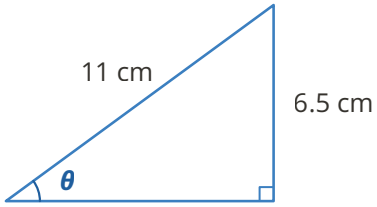
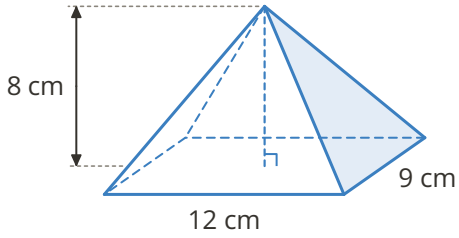
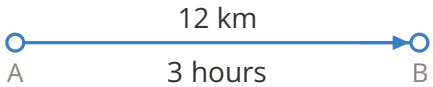
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 2 · Session 1

TERM 4 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find θ, correct to the nearest degree. 	FORMULA SUBSTITUTE ANSWER
2 Find the volume of this pyramid. 	FORMULA $V = \frac{1}{3}Ah$ (A = area of the base) COPY SUBSTITUTE ANSWER
3 Find the original price of the shirt. Discount 20% Sale price $48	FORMULA SUBSTITUTE ANSWER
4 A walker travels from A to B. Find the average speed. 	FORMULA SUBSTITUTE ANSWER
5 Find the probability that it does not rain. $P(\text{rain}) = 0.4$	FORMULA SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

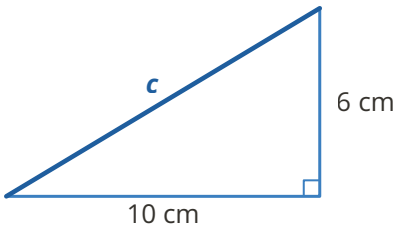
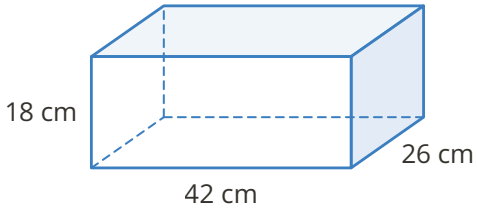
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 2 · Session 2

TERM 4 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find c, correct to 1 decimal place.  	FORMULA SUBSTITUTE ANSWER
2 How many litres does this tank hold, correct to 1 decimal place?  	FORMULA SUBSTITUTE ANSWER
3 Find Isla's commission. Value of sales: $9500 Commission rate: 6%	FORMULA SUBSTITUTE ANSWER
4 Share $95 in the ratio 3 : 2. Give both shares.	FORMULA SUBSTITUTE ANSWER
5 A spinner was spun 50 times. Find the relative frequency of red. Colour: Red, Blue, Green Frequency: 14, 26, 10	FORMULA SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

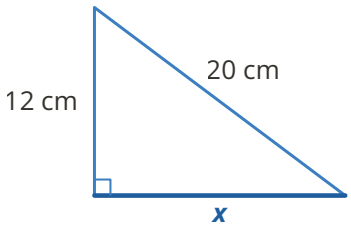
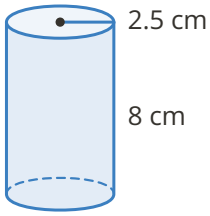
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 2 · Session 3

TERM 4 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA SUBSTITUTE ANSWER
2 Find the volume of this cylinder, correct to 1 decimal place.  	FORMULA SUBSTITUTE ANSWER
3 Find the new price. Original price: $64.50 Increase: 12%	FORMULA SUBSTITUTE ANSWER
4 Write as a single power. $2^3 \times 2^4$	FORMULA SUBSTITUTE ANSWER
5 Find the value of y when $x = 5$. $y = 2x - 3$	FORMULA SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

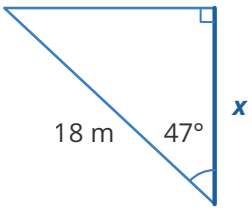
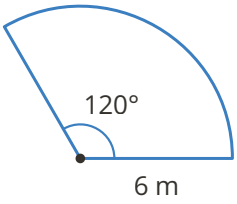
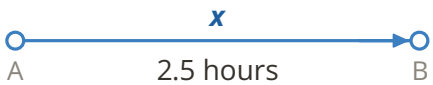
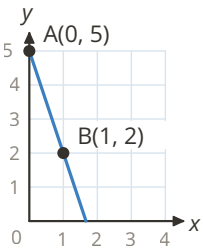
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 3 · Session 1

TERM 4 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find x, correct to 1 decimal place. 	FORMULA SUBSTITUTE ANSWER
2 Find the area of this sector, correct to 1 decimal place. 	FORMULA SUBSTITUTE ANSWER
3 Find the simple interest. Amount invested: $1200 Interest rate: 4.5% per year Time: 3 years	FORMULA SUBSTITUTE ANSWER
4 A car travels from A to B at an average speed of 63 km/h. Find x. 	FORMULA SUBSTITUTE ANSWER
5 Find the gradient of line AB. 	FORMULA SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

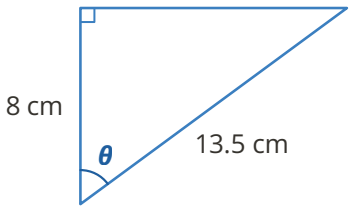
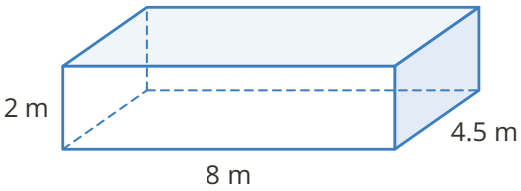
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 3 · Session 2

TERM 4 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find θ, correct to the nearest degree.  	FORMULA SUBSTITUTE ANSWER
2 Find the surface area of this prism.  	FORMULA SUBSTITUTE ANSWER
3 Find Finn's total holiday pay. Weekly pay: $1350 Holiday: 4 weeks Leave loading: 17.5%	FORMULA SUBSTITUTE ANSWER
4 Write in scientific notation. 45 000	FORMULA SUBSTITUTE ANSWER
5 One marble is chosen at random from this bag. Find the probability that it is black. Colour: Black, White Marbles: 7, 3	FORMULA SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

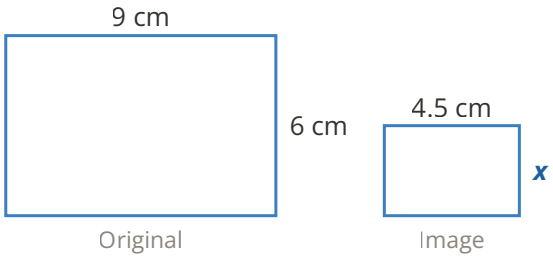
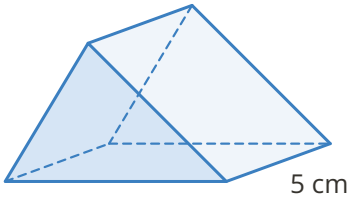
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 3 · Session 3

TERM 4 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 These rectangles are similar. Find x.  	FORMULA SUBSTITUTE ANSWER
2 The shaded face has an area of 20 cm^2. Find the volume of the prism.  	FORMULA SUBSTITUTE ANSWER
3 Find Mia's total pay for the week. Weekly base wage: \$600 Commission: 4% of sales Sales this week: \$9375	FORMULA SUBSTITUTE ANSWER
4 Write with a positive index. 3^{-4}	FORMULA SUBSTITUTE ANSWER
5 Find the probability that the target is not hit. $P(\text{hit}) = \frac{7}{9}$	FORMULA SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

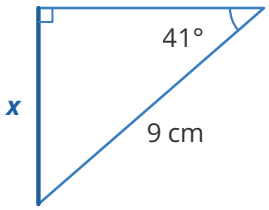
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 4 · Session 1

TERM 4 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find x, correct to 1 decimal place. 	FORMULA SUBSTITUTE ANSWER
2 Find the surface area of this closed cylinder, correct to 1 decimal place. 	FORMULA SUBSTITUTE ANSWER
3 Find the number. 15% of a number is 45	FORMULA SUBSTITUTE ANSWER
4 Share 2.4 kg in the ratio 1 : 2. Give both shares.	FORMULA SUBSTITUTE ANSWER
5 On how many days would you expect the bus to be late? P(the bus is late) 0.35 School days 400	FORMULA SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

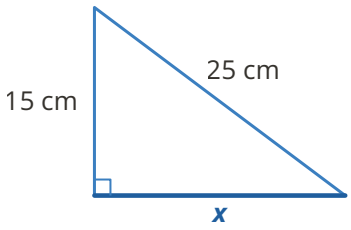
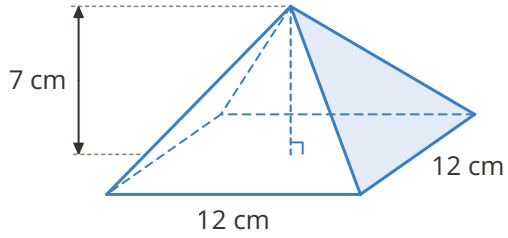
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 4 · Session 2

TERM 4 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA SUBSTITUTE ANSWER
2 Find the volume of this pyramid.  	FORMULA $V = \frac{1}{3}Ah$ (A = area of the base) COPY SUBSTITUTE ANSWER
3 Hamish earns a salary of \$58 900 a year. Find Hamish's fortnightly pay, to the nearest cent.	FORMULA SUBSTITUTE ANSWER
4 Simplify. $y^9 \div y^4$	FORMULA SUBSTITUTE ANSWER
5 State the gradient and the y-intercept. $y = 3x - 2$	FORMULA SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

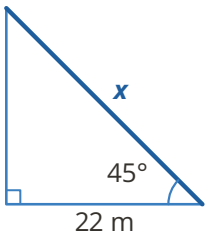
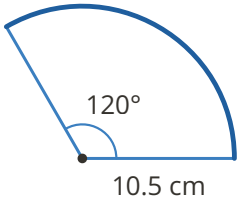
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 4 · Session 3

TERM 4 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find x, correct to 1 decimal place. 	FORMULA SUBSTITUTE ANSWER
2 Find the arc length, correct to 1 decimal place. 	FORMULA SUBSTITUTE ANSWER
3 Find the simple interest. Amount invested: $1500 Interest rate: 6% per year Time: 4 years	FORMULA SUBSTITUTE ANSWER
4 Find x when $y = 11$, $m = 4$ and $c = -3$. $y = mx + c$	FORMULA SUBSTITUTE ANSWER
5 Find the mean of these goal tallies. Goals: 7, 9, 4, 10, 5	FORMULA SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

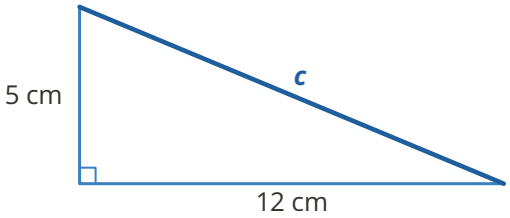
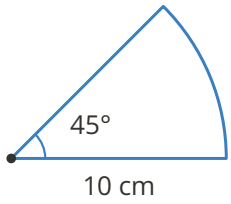
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 5 · Session 1

TERM 4 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the length of the hypotenuse, c.  	FORMULA SUBSTITUTE ANSWER
2 Find the area of this sector, correct to 1 decimal place.  	FORMULA SUBSTITUTE ANSWER
3 Find the percentage loss. Cost price: $45 Selling price: $36	FORMULA SUBSTITUTE ANSWER
4 Write in scientific notation. 28 500 000	FORMULA SUBSTITUTE ANSWER
5 Find the probability that the flight is not delayed. $P(\text{delay}) = 0.125$	FORMULA SUBSTITUTE ANSWER

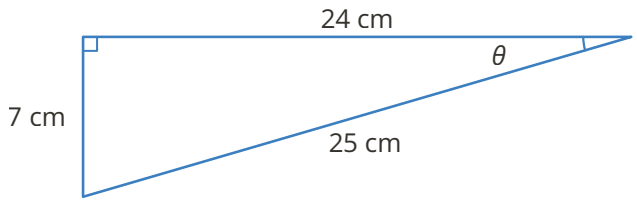
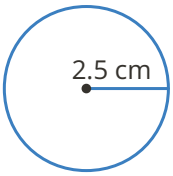
KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

Week 5 · Session 2

TERM 4 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Write sin θ as a fraction.  	FORMULA SUBSTITUTE ANSWER
2 Find the area of this circle, correct to 1 decimal place.  	FORMULA SUBSTITUTE ANSWER
3 Find the sale price, to the nearest cent. Original price: $59.99 Discount: 35%	FORMULA SUBSTITUTE ANSWER
4 Evaluate. 2^{-3}	FORMULA SUBSTITUTE ANSWER
5 Heads came up 42 times in 100 tosses. Find the relative frequency of heads.	FORMULA SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

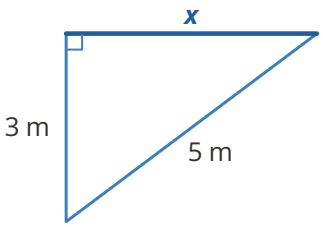
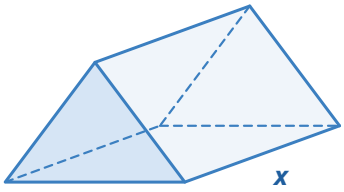
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 5 · Session 3

TERM 4 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA SUBSTITUTE ANSWER
2 The shaded face has an area of 12 cm^2 and the volume is 96 cm^3. Find x.  	FORMULA SUBSTITUTE ANSWER
3 Find Sienna's total pay for the week. Hourly rate: $29.90 Normal hours: 36 hours Time-and-a-half: 3 hours	FORMULA SUBSTITUTE ANSWER
4 Simplify. $(5x^4)^2$	FORMULA SUBSTITUTE ANSWER
5 State the gradient and the y-intercept. $y = 0.5x - 3$	FORMULA SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

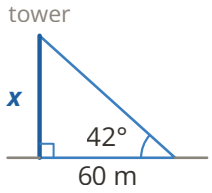
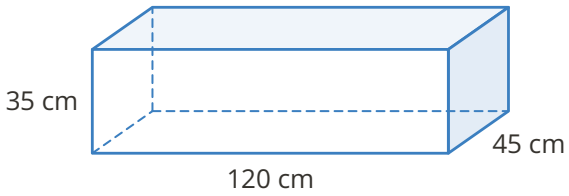
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 6 · Session 1

TERM 4 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 The angle of elevation is 42°. Find x, correct to 1 decimal place. 	FORMULA SUBSTITUTE ANSWER
2 How many litres does this tank hold? 	FORMULA SUBSTITUTE ANSWER
3 Find Raj's total holiday pay. Weekly pay: $1427 Holiday: 4 weeks Leave loading: 17.5%	FORMULA SUBSTITUTE ANSWER
4 Find h when $A = 27$ and $b = 12$. $A = \frac{1}{2}bh$	FORMULA SUBSTITUTE ANSWER
5 Find the mean of these scores, correct to 1 decimal place. Score: 1, 2, 3, 4, 5 Frequency: 3, 6, 8, 5, 2	FORMULA SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

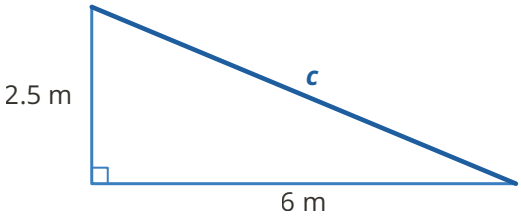
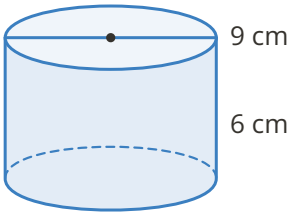
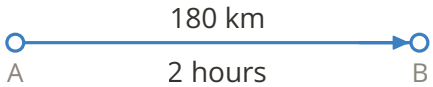
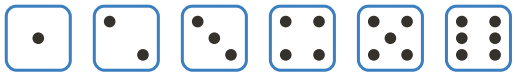
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 6 · Session 2

TERM 4 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the length of the hypotenuse, c. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____
2 Find the surface area of this closed cylinder, correct to 1 decimal place. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____
3 Find the new population of the town. Population 3840 Increase 2.5%	FORMULA _____ SUBSTITUTE _____ ANSWER _____
4 A car travels from A to B. Find the average speed. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____
5 A fair die is rolled. Find the probability of rolling a prime number. 	FORMULA _____ SUBSTITUTE _____ ANSWER _____

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

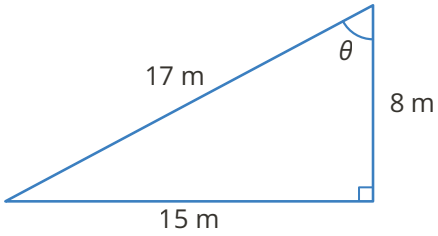
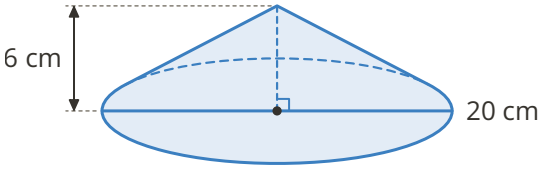
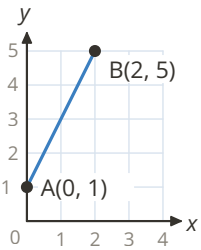
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 6 · Session 3

TERM 4 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Write $\tan\theta$ as a fraction.  	FORMULA SUBSTITUTE ANSWER
2 Find the volume of this cone, correct to 1 decimal place.  	FORMULA $V = \frac{1}{3}\pi r^2 h$ COPY SUBSTITUTE ANSWER
3 Find the original price of the cap. Discount 35% Sale price $33.15	FORMULA SUBSTITUTE ANSWER
4 Share $250 in the ratio 3 : 5. Give both shares.	FORMULA SUBSTITUTE ANSWER
5 Find the gradient of line AB.  	FORMULA SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

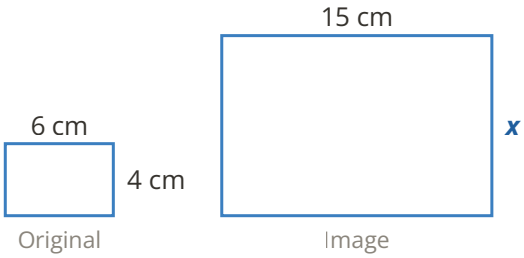
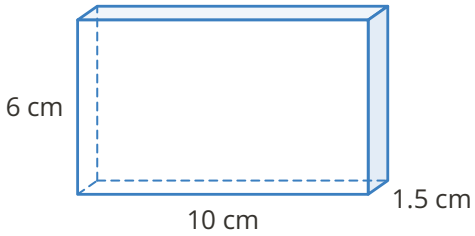
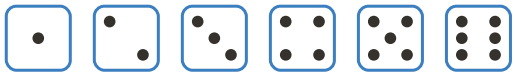
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 7 · Session 1

TERM 4 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 These rectangles are similar. Find x.  	FORMULA SUBSTITUTE ANSWER
2 Find the surface area of this prism.  	FORMULA SUBSTITUTE ANSWER
3 Find Arjun's total pay for the week. Hourly rate: $26.75 Normal hours: 30 hours Double time: 4 hours	FORMULA SUBSTITUTE ANSWER
4 Write in scientific notation. 0.00045	FORMULA SUBSTITUTE ANSWER
5 A fair die is rolled 450 times. How many times would you expect a 1 or a 2? 	FORMULA SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

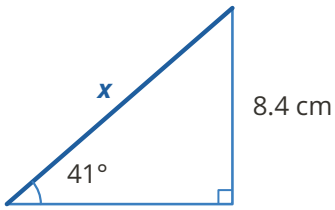
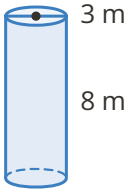
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 7 · Session 2

TERM 4 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find x, correct to 1 decimal place. 	FORMULA SUBSTITUTE ANSWER
2 Find the volume of this cylinder, correct to 1 decimal place. 	FORMULA SUBSTITUTE ANSWER
3 Find the percentage loss. Cost price \$120 Selling price \$90	FORMULA SUBSTITUTE ANSWER
4 Simplify. $3x^4 \times 2x^3$	FORMULA SUBSTITUTE ANSWER
5 Find the value of y when $x = 3$. $y = -2x + 10$	FORMULA SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

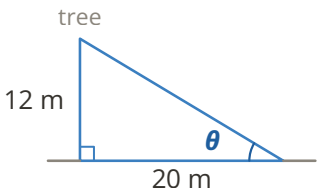
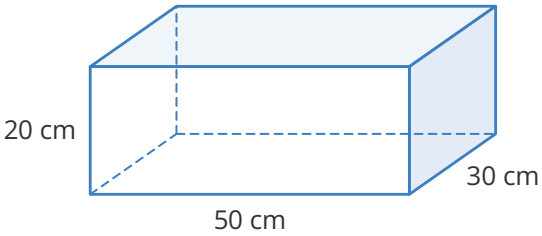
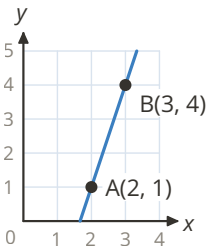
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 7 · Session 3

TERM 4 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the angle of elevation, θ, correct to the nearest degree. 	FORMULA SUBSTITUTE ANSWER
2 How many litres does this tank hold? 	FORMULA SUBSTITUTE ANSWER
3 Find Leo's total pay for the week. Weekly base wage: $520 Commission: 2.5% of sales Sales this week: $18 400	FORMULA SUBSTITUTE ANSWER
4 Find C when $d = 10$. $C = 5 + 2.4d$	FORMULA SUBSTITUTE ANSWER
5 Find the gradient of line AB. 	FORMULA SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

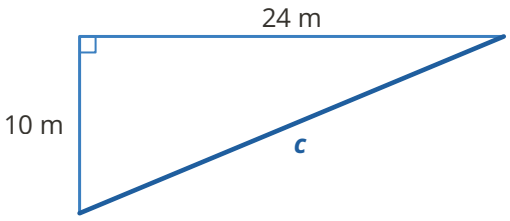
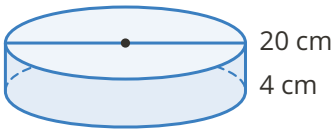
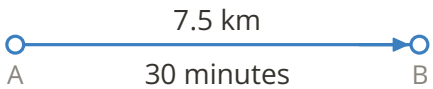
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 8 · Session 1

TERM 4 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the length of the hypotenuse, c. 	FORMULA _____ SUBSTITUTE _____ _____ _____ ANSWER _____
2 Find the surface area of this closed cylinder, correct to 1 decimal place. 	FORMULA _____ SUBSTITUTE _____ _____ _____ ANSWER _____
3 Find Ali's total holiday pay. Weekly pay: $1100 Holiday: 4 weeks Leave loading: 17.5%	FORMULA _____ SUBSTITUTE _____ _____ _____ ANSWER _____
4 A cyclist travels from A to B. Find the average speed in km/h. 	FORMULA _____ SUBSTITUTE _____ _____ _____ ANSWER _____
5 Find the mean of these scores. Scores: 30, 25, 35, 20, 40	FORMULA _____ SUBSTITUTE _____ _____ _____ ANSWER _____

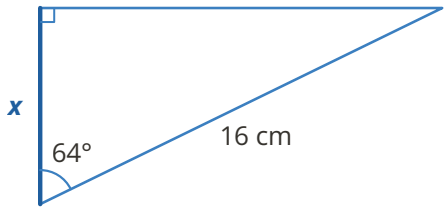
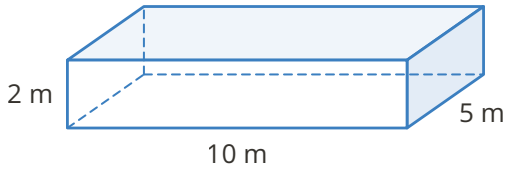
KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

Week 8 · Session 2

TERM 4 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find x, correct to 1 decimal place.  	FORMULA SUBSTITUTE ANSWER
2 Find the surface area of this prism.  	FORMULA SUBSTITUTE ANSWER
3 Find the simple interest. Amount invested: $12 000 Interest rate: 2% per year Time: 5 years	FORMULA SUBSTITUTE ANSWER
4 Write with a positive index. 5^{-2}	FORMULA SUBSTITUTE ANSWER
5 A letter is chosen at random from this word. Find $P(\text{B})$. PROBABILITY	FORMULA SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

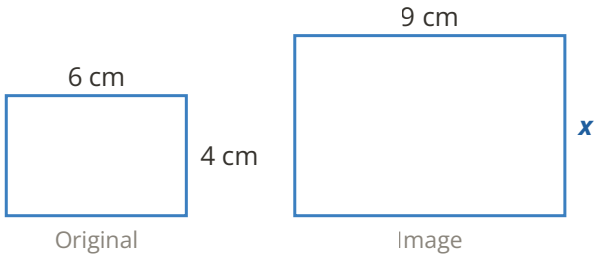
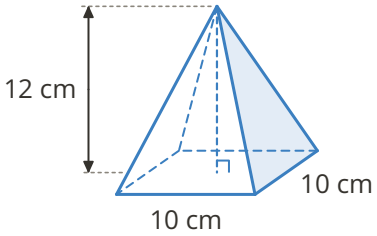
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 8 · Session 3

TERM 4 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 These rectangles are similar. Find x.  	FORMULA SUBSTITUTE ANSWER
2 Find the volume of this pyramid.  	FORMULA $V = \frac{1}{3}Ah$ (A = area of the base) COPY SUBSTITUTE ANSWER
3 Mia earns a salary of \$78 000 a year. Find Mia's weekly pay.	FORMULA SUBSTITUTE ANSWER
4 Share \$100 in the ratio 3 : 5. Give both shares.	FORMULA SUBSTITUTE ANSWER
5 Find the probability that the marble is not blue. $P(\text{blue}) = 0.36$	FORMULA SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

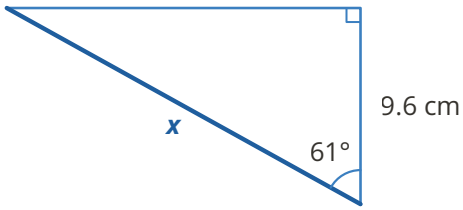
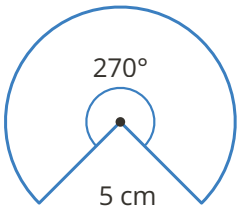
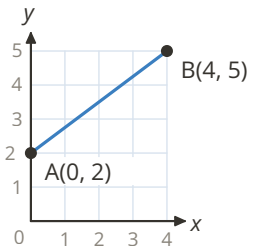
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 9 · Session 1

TERM 4 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find x, correct to 1 decimal place. 	FORMULA SUBSTITUTE ANSWER
2 Find the area of this sector, correct to 1 decimal place. 	FORMULA SUBSTITUTE ANSWER
3 Find Ivy's total holiday pay. Weekly pay: $1098.40 Holiday: 4 weeks Leave loading: 17.5%	FORMULA SUBSTITUTE ANSWER
4 Find C when $F = 95$. $F = 1.8C + 32$	FORMULA SUBSTITUTE ANSWER
5 Find the gradient of line AB. 	FORMULA SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

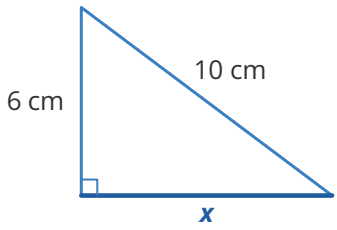
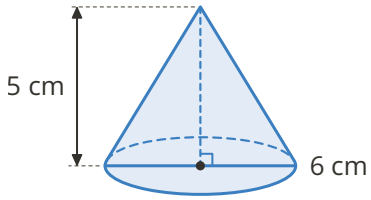
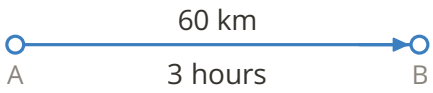
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 9 · Session 2

TERM 4 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA SUBSTITUTE ANSWER
2 Find the volume of this cone, correct to 1 decimal place.  	FORMULA $V = \frac{1}{3}\pi r^2 h$ COPY SUBSTITUTE ANSWER
3 Find the number. 40% of a number is 36	FORMULA SUBSTITUTE ANSWER
4 A cyclist travels from A to B. Find the average speed.  	FORMULA SUBSTITUTE ANSWER
5 Find the mean of these scores. Score: 1, 2, 3, 4, 5 Frequency: 1, 1, 2, 4, 7	FORMULA SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

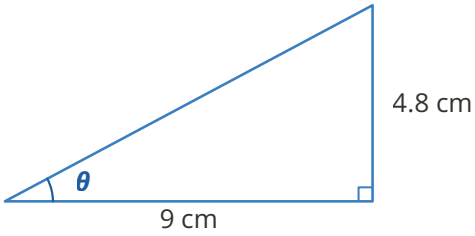
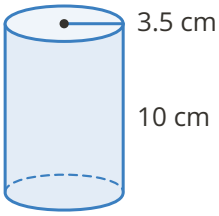
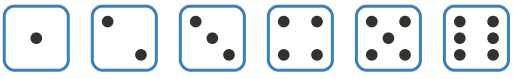
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 9 · Session 3

TERM 4 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find θ, correct to the nearest degree.  	FORMULA SUBSTITUTE ANSWER
2 Find the volume of this cylinder, correct to 1 decimal place.  	FORMULA SUBSTITUTE ANSWER
3 Find the new amount. Decrease \$1250 by 7.5%	FORMULA SUBSTITUTE ANSWER
4 Evaluate. $2^{-1} + 2^{0}$	FORMULA SUBSTITUTE ANSWER
5 A fair die is rolled. Find the probability of rolling a number greater than 1. 	FORMULA SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

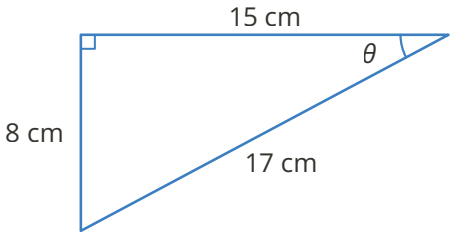
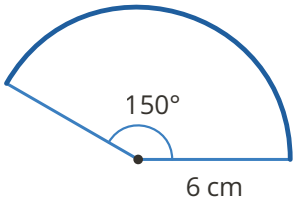
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 10 · Session 1

TERM 4 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Write sin θ as a fraction.  	FORMULA SUBSTITUTE ANSWER
2 Find the arc length, correct to 1 decimal place.  	FORMULA SUBSTITUTE ANSWER
3 Find the simple interest on this loan. Amount borrowed: $8000 Interest rate: 6% per year Time: 9 months	FORMULA SUBSTITUTE ANSWER
4 Share \$120 in the ratio 3 : 5. Give both shares.	FORMULA SUBSTITUTE ANSWER
5 Find the probability that the counter is not red. $P(\text{red}) = \frac{1}{4}$	FORMULA SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

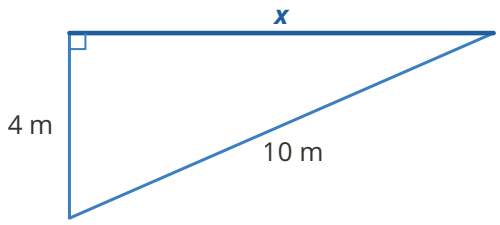
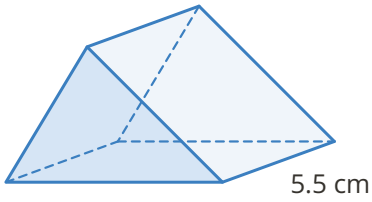
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 10 · Session 2

TERM 4 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find x, correct to 1 decimal place. 	FORMULA SUBSTITUTE ANSWER
2 The shaded face has an area of 20 cm^2. Find the volume of the prism. 	FORMULA SUBSTITUTE ANSWER
3 Find the percentage profit. Cost price: $12.50 Selling price: $15	FORMULA SUBSTITUTE ANSWER
4 Simplify. $24p^9 \div 8p^4$	FORMULA SUBSTITUTE ANSWER
5 A six came up 33 times in 150 rolls. Find the relative frequency of a six.	FORMULA SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

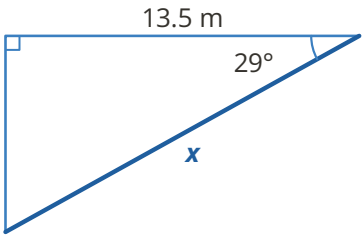
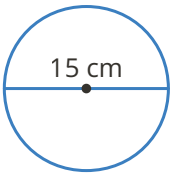
MATHEMATICS · YEAR 10 · THE FRACTAL FIVE

Week 10 · Session 3

TERM 4 Write the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find x, correct to 1 decimal place.  	FORMULA SUBSTITUTE ANSWER
2 Find the area of this circle, correct to 1 decimal place.  	FORMULA SUBSTITUTE ANSWER
3 Find Chloe's commission. Value of sales: $25 000 Commission rate: 4%	FORMULA SUBSTITUTE ANSWER
4 Write as a basic numeral. 5×10^3	FORMULA SUBSTITUTE ANSWER
5 Find the value of y when $x = 4$. $y = 3x + 1$	FORMULA SUBSTITUTE ANSWER